

Methods

Individual Neural Efficiency Regressions

To characterise individual-level NE learning trajectories, simple OLS regressions ($NE \sim \text{sessao_num}$) were estimated independently for each participant-ROI combination ($N = 9$ data points per regression, training sessions 1-9). The OLS slope represents the rate of NE change per session. Participants were classified according to the significance of their individual slope ($p < .05$, $p < .10$, or n.s.). Between-group differences in slope distributions were assessed using independent-samples t-tests on mean slope magnitude and Fisher exact tests on the proportion of participants with significant slopes, conducted separately for each ROI.

Mann-Whitney Group Comparison

To assess overall group differences in NE magnitude, mean NE was computed per participant for each ROI (averaged across training sessions 1-9). Mann-Whitney U tests (two-tailed) compared the Accelerated and Non-Accelerated group distributions. Effect size was quantified as the rank-biserial correlation $r = 1 - (2U / n1*n2)$, classified as small ($r < 0.3$), medium ($0.3 \leq r < 0.5$), or large ($r \geq 0.5$).

Results

Individual NE Regressions

In the Frontal ROI, the Accelerated group showed a mean positive NE slope of $M = 0.113$ ($SD = 0.075$), with 1 of 7 participants reaching significance ($p < .05$). The Non-Accelerated group mean slope was $M = 0.046$ ($SD = 0.107$), with 1 of 7 participants significant. Between-group comparisons were non-significant (t-test: $p = .198$; Fisher exact: $p = 1.0$).

In the Temporal ROI, the Accelerated group showed a mean slope of $M = 0.136$ ($SD = 0.109$), with 3 of 7 participants showing significant positive slopes ($p < .05$). The Non-Accelerated group mean slope was $M = 0.027$ ($SD = 0.109$), with 2 of 7 significant. The between-group difference approached but did not reach statistical significance (t-test: $p = .086$; Fisher exact: $p = 1.0$).

Mann-Whitney Group Comparison of Mean NE

When mean NE across training sessions was compared between groups, the Accelerated group demonstrated significantly higher NE than the Non-Accelerated group in both ROIs.

Frontal ROI: Accelerated $M = 0.470$ ($SD = 0.286$) vs Non-Accelerated $M = -0.470$ ($SD = 0.762$); Mann-Whitney $U = 46$, $p = .004$, $r = -0.878$ (large effect size).

Temporal ROI: Accelerated $M = 0.543$ ($SD = 0.482$) vs Non-Accelerated $M = -0.543$ ($SD = 0.725$); Mann-Whitney $U = 47$, $p = .002$, $r = -0.918$ (large effect size).

These results indicate that, across the full training period, the Accelerated group maintained substantially higher neural efficiency in both frontal and temporal regions, with large effect sizes in both cases.